

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Aquaculture - Set 1

Mark scheme

AQA spec 3.5.2.3; Biological resources; textbook pp. 352-359

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• MEDCs: market demand and higher prices for the flavour of carnivorous fish (salmon, trout) justify high inputs (fishmeal, cooler climates suit them) • LEDCs: herbivores / omnivores feed on natural vegetation, phytoplankton and wastes so need low inputs; warm climates suit tilapia • Herbivores are more energy-efficient (lower trophic level) so more food per unit area	3
01.2	• Some genetically female (XX) fish are treated with testosterone so they become functional males producing only X-carrying sperm • Crossed with normal females (XX) all offspring are XX females • Female trout meat has a better flavour; in tilapia the reverse (all males) because males grow larger / more efficiently	3
01.3	• Salmon need high oxygen: water sprays / weirs aerate; wastes removed to prevent deoxygenation • Constant circular flow keeps fish swimming the same way - fewer collisions / injuries, more muscular fish, higher stocking density • Artificial lighting delays maturation (salmon stop growing at maturity) so larger fish; controls smoltification • Cool sites (Scotland, Norway) because warm water holds less oxygen although warmth increases growth	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 352-355; AQA spec 3.5.2.3.	
02.1	• Extensive: low inputs of energy, food and chemicals so low cost and low environmental impact; sustainable; but low yield per area and dependent on natural productivity • Intensive: control of all limiting factors gives very high yield and profit per area; but high inputs (fishmeal from wild fish, energy), disease and lice at high density, pollution from wastes, antibiotic / pesticide use, escapes • Judgement depends on species, market and site	4

Question	Marking guidance	Marks
02.2	- Adults with desirable characteristics (fast growth, shape, colour) stripped of eggs and milt which are mixed for fertilisation - Eggs raised in aerated freshwater tanks, dead / diseased eggs removed daily; fry fed fishmeal pellets and moved through tanks as they grow - After 12-18 months smolts transferred to seawater cages, fed pellets until harvest at 3-5 kg	3
02.3	- About 10% of the world's mangroves have been cleared for coastal shrimp lagoons (up to 100 ha), removing nurseries and coastal protection - Lagoons need water exchange to remove wastes and aeration to prevent deoxygenation as wastes decompose - Eyestalk ablation stimulates breeding; high densities increase disease so antibiotics / pesticides used	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 352-356; AQA spec 3.5.2.3.	
03.1	- Carnivores fed pellets of sandeels, herring, anchovy; overfishing sandeels reduces puffins and terns; reduced by more plant protein / vegetable oil in pellets or farming herbivores (tilapia) - High density spreads lice (which spread to wild fish) and infections; pesticides (organophosphates, pyrethroids, hydrogen peroxide) kill wildlife; antibiotics breed resistant bacteria; reduced by lower densities, single-age tanks with flow from young to old, wrasse as biological control, warm-water washes, circulating flow - Escaped farmed fish breed with wild fish introducing disadvantageous genes (bright colour), or non-indigenous escapees colonise as predators / competitors; reduced by better cage design, sterile triploid fish (pressure-treated eggs), species that cannot breed locally (Japanese oysters in UK) - Faeces and uneaten food decompose causing deoxygenation under cages; reduced by monitored feeding, siting cages in strong currents, effluent treatment / reed beds for freshwater farms, IMTA with seaweed and shellfish absorbing wastes	8
03.2	- Predators are killed under licence to protect stock, reducing wild populations of protected / valued species - Better cage designs, netting, fencing, bird scarers exclude predators without killing	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 356-359; AQA spec 3.5.2.3.	
04.1	- Silver carp eat phytoplankton, bighead carp zooplankton, grass carp and tilapia vegetation, common carp bottom-feed on molluscs / insects, catfish or perch eat small fish - Species occupy different niches / trophic levels so they do not compete and all food sources are used - Bottom feeders re-suspend nutrients boosting plankton / plant growth - Predators control over-breeding species (tilapia) so remaining fish grow larger; total yield exceeds any single species	4

Question	Marking guidance	Marks
04.2	• IMTA: fed species (salmon, shrimp) alongside inorganic extractive species (seaweeds absorbing ammonia and phosphate) and organic extractive species (filter-feeding mussels eating waste particles / plankton) • Wastes of one species become nutrients for another so less pollution and extra harvests • Aquaponics: aquaculture water rich in nutrients / organic matter feeds hydroponic salad crops (lettuce, herbs, watercress) • Plants remove the nutrients that would cause deoxygenation and cleaned water returns to the fish (tilapia, carp, catfish); rice-fish paddies similar	4
04.3	• Bluefin tuna do not breed well in captivity so young fish are caught from the wild and fattened in sea cages • Production still depends on (and removes juveniles from) the wild population; carnivores need large fish inputs	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 357-358; AQA spec 3.5.2.3.	
05.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Context: wild catches have not increased since about 1990 as stocks are fully or over-exploited; aquaculture output growing rapidly and approaching the wild catch; fish are the last large-scale hunted food • Advantages of aquaculture: control of limiting factors, selective breeding, gender control, high yields per area; fish have low FCRs (no thermoregulation / support) compared with livestock; herbivores (tilapia, carp) and molluscs very efficient; polyculture, IMTA, aquaponics, rice-fish • Limits: carnivorous species (salmon, shrimp) need fishmeal from wild fish so still depend on fishing and deplete forage fish (sandeels - puffins); some species (bluefin tuna) will not breed in captivity; consumer preference for carnivores • Environmental impacts: habitat loss (mangroves), organic pollution, pesticides, antibiotics and resistance, lice, escapes and gene pool contamination, predator control; and how they can be reduced • Fishing can be sustainable with quotas, gear design, NTZs, size limits, effort limits; MSY management needs good data • Judgement: aquaculture can supply much of the growth in demand, especially herbivores and shellfish, but cannot fully replace fishing; both need sustainable management	9
05.2	• A species that is given artificial food (pellets) rather than relying on natural food in the water, eg salmon, shrimp	1

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 338-359; AQA spec 3.5.2.	